Artifical Intelligence Basics Course Summary

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1 Key Terms

- **Turing Test**: An operational test for intelligent behavior, where a human evaluator engages in natural language conversations with both a human and a machine, attempting to distinguish them. The machine passes if the evaluator cannot reliably identify it as such.
- **Chinese Room Argument**: A thought experiment that challenges the idea that passing the Turing Test implies true understanding or consciousness.
- **Minimax Search**: A decision rule used in game theory and search algorithms to find the optimal move for a player, assuming the opponent also plays optimally.
- **Alpha-Beta Pruning**: A search algorithm optimization technique that reduces the search space in Minimax search by eliminating branches that cannot affect the optimal decision.
- **Perceptron**: A fundamental unit in artificial neural networks that performs a weighted sum of its inputs and applies an activation function to produce an output.
- **Reinforcement Learning**: A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties for its actions.

2 Problem Types

- **Understanding the Nature of Intelligence**: This problem explores the definition and characteristics of intelligence, both human and artificial. Different perspectives and definitions of AI are examined.
- **Evaluating Intelligent Behavior**: This problem focuses on developing methods for assessing whether a machine exhibits intelligent behavior. The Turing Test and its limitations are discussed, along with counterarguments like Searle's Chinese Room argument.
- **Overcoming the Limits of Search**: This problem addresses the computational challenges of exhaustive search in complex games like chess and Go. The exponential growth of search space is highlighted, leading to the need for machine learning techniques to supplement search.
- **Addressing AI Safety and Ethical Concerns**: This problem explores the potential risks and ethical implications of AI systems. Examples include bias in algorithms and the vulnerability of neural networks to adversarial attacks.
- **Bridging the Gap Between Different AI Domains**: This problem investigates the commonalities and differences between seemingly disparate AI tasks, such as game playing and speech synthesis. The shared algorithmic components and challenges are analyzed.
- **Understanding the Challenges of Real-World AI Applications**: This problem examines the difficulties of applying AI to complex real-world scenarios, such as autonomous driving. The reasons for choosing autonomous vehicles as an AI Grand Challenge are explored.

3 Algorithm Solutions

- **Minimax Search**: A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning**: An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.